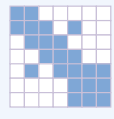
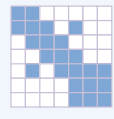
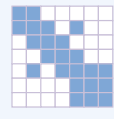


Mandala Workflow Pipeline

Learning sparse electronic-structure operators with $E(3)$ -equivariant graph neural networks

1 Backend data

OpenMX • FHI-aims • PySCF



H S D

Electronic-structure outputs
(Hamiltonian, overlap, density matrices)

2 Snapshots

`Snapshot.from_openmx()`
 $\{L, R, Z\}$
and $\{H, S, D, E_b, N_e\}$



Lattice vectors, positions,
atomic species, Hamiltonian,
overlap and density matrices,
band energy,
number of electrons

3 Dataset

`DatasetFactory.create()`

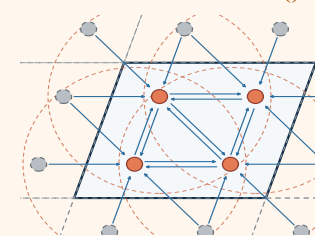
`E3GNNDataset`



Graph samples with irreps,
blocks, features, and targets

4 $E(3)$ -Equivariant GNNs

`E3GNN.forward()`



H

Hamiltonian
head

S

Overlap
head

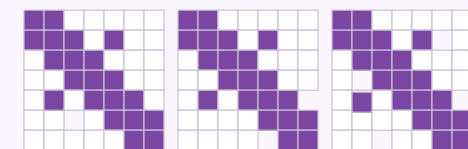
D

Density
head

$E(3)$ -equivariant message passing,
tensor products, and readout heads

5 Predictions

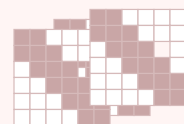
H, S, D



Sparse operator blocks
in symmetry-adapted bases

A Training / validation

`E3GNN.training_step()`
matrix + observable losses



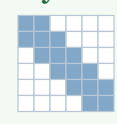
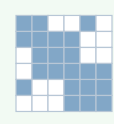
+

E_b ϵ_i N_e

Observable-guided supervision with
per-target weighting and schedules

B Postprocessing

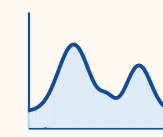
symmetrization • gauge correction
density normalization



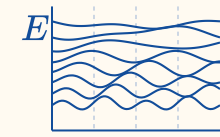
Physics-aware refinement of
 H, S, D

C Analysis

metrics • DOS • band structure



DOS



k

Quantitative evaluation and
electronic-structure insights